

PROJECT TITLE
BIM-Based Residential Building Modelling &
Documentation

Role

BIM Modeler | Architectural & Structural Coordination

Software

Autodesk Revit

Project Type

Multi-Storied Residential Building

Prepared by

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Diploma in Civil Engineering

01. PROJECT OVERVIEW

This project demonstrates the **end-to-end BIM workflow** followed for planning, modelling, coordination, and documentation of a **multi-storied residential building** using **Autodesk Revit**.

The focus was on:

- Accurate 3D modelling
- Regulation-compliant planning
- Automated drawing extraction
- Industry-standard documentation

The deliverable mirrors how BIM is executed in **EPC companies, design consultancies, and MNC construction firms**.

02. PROJECT OBJECTIVE

The objective of this project was to create a **single source of truth BIM model** that integrates architectural planning, structural elements, and documentation outputs, minimizing drafting errors and ensuring coordinated design.

03. SCOPE OF WORK (BIM RESPONSIBILITY MATRIX)

Architectural BIM Scope

- Floor plans (Ground & upper floors)
- Door, window, and room layouts
- Staircase and circulation planning
- Roof and terrace modelling

Structural BIM Scope

- RCC slabs, beams, and columns
- Staircase structural components
- Tie beams and level coordination

Documentation Scope

- Plans, elevations, and sections
- Door & window schedules
- Area and level consistency checks

04. BIM MODELLING APPROACH

A **Level-based BIM workflow** was followed:

- Project base point and levels established
- Grids aligned for architectural and structural coordination
- Parametric families used for doors, windows, slabs, and stairs
- Model accuracy maintained to ensure drawing consistency

All drawings were **live-linked to the model**, eliminating manual drafting errors.

05. BIM EXECUTION WORKFLOW (Dedicated Page)

BIM Execution Strategy

Step 1 – Project Setup

- Levels, grids, and project units defined
- Naming conventions standardized

Step 2 – Architectural Modelling

- Walls, floors, roofs, doors, and windows modelled
- Space planning optimized for ventilation and usability

Step 3 – Structural Modelling

- RCC slabs and columns placed with level coordination
- Stair structural alignment checked through sections

Step 4 – Model Coordination

- Clash-free alignment between architectural and structural elements
- Vertical services alignment across floors

Step 5 – Drawing & Schedule Extraction

- Plans, elevations, and sections auto-generated
- Door and window schedules extracted directly from the model

This workflow reflects **real BIM execution plans (BEP)** followed in professional projects.

06. KEY BIM DELIVERABLES

- Ground Floor Plan
- Upper Floor Plans
- Roof Plan
- Front Elevation
- Building Sections
- Staircase Section
- Door & Window Schedule

All deliverables were **generated from a single coordinated BIM model.**

07. STANDARDS & COMPLIANCE

- Planning parameters aligned with municipal norms
 - Structural elements modelled referencing IS & NBC guidelines
 - Floor heights, stair dimensions, and slab thickness maintained as per standards
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08. TOOLS & TECHNOLOGY STACK

- Autodesk Revit – Primary BIM authoring
 - Parametric families for scalable design
 - Sheet sets and annotations managed within Revit
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09. VALUE DELIVERED (WHY THIS PROJECT MATTERS)

- Reduced drawing rework through BIM coordination
 - Improved design clarity using 3D visualization
 - Faster documentation turnaround
 - Portfolio demonstrates **job-ready BIM capability**, not just software knowledge.
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10. SKILLS DEMONSTRATED

- BIM Modelling (Architecture & Structure)
- Revit Documentation Workflow
- Residential Building Planning
- Drawing Coordination & Sheet Management
- Industry-aligned execution mindset

11. CONCLUSION

This project showcases my ability to **execute residential building projects using BIM methodology**, aligning with the expectations of **EPC firms, design consultancies, and MNC construction companies**.

The workflow, documentation quality, and coordination reflect **entry-level BIM Engineer / Revit Modeler readiness**.

— BIM PORTFOLIO CASE STUDY —

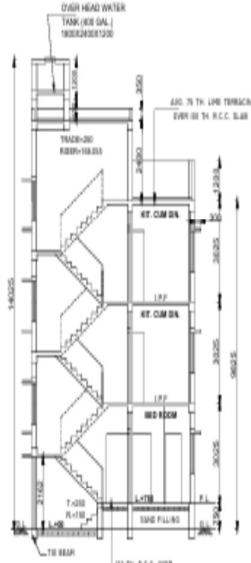
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FRONT ELEVATION
SCALE=1:100

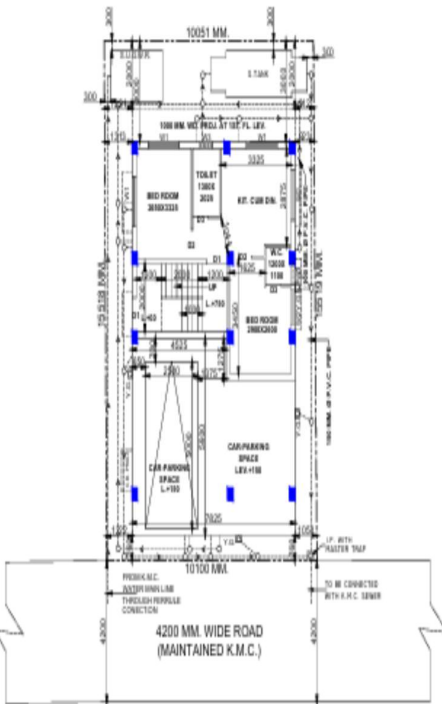


SECTION AT A-A'
SCALE=1:100



SECTION AT B-B'
SCALE=1:100

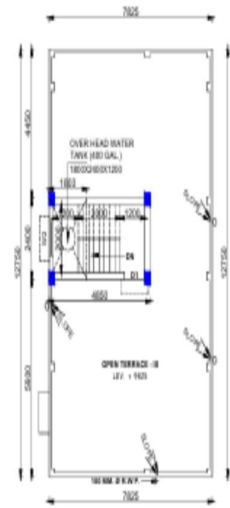
THE DEPTH OF SEPTIC TANK & SUMP, SHOULD NOT EXCEED THE DEPTH OF FOUNDATION OF THE BUILDING



GROUND FLOOR PLAN
SCALE=1:100



1ST. & 2ND. FLOOR PLAN
SCALE=1:100



ROOF PLAN
SCALE=1:100